



UNIVERSITY OF MINNESOTA

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Elevator Statement

- Our team is developing a portable device that assesses the viability of bowel tissue for mysenteric ischemia

- The goal is to increase the accuracy of quantifying bowel viability through the characterization of electrical impedance and its relation to metabolic activity.

Clinical Need

Mesenteric ischemia is a life-threatening condition that affects 2-3 out of 100,000 patients annually [1]. It is caused by insufficient blood flow to the intestines, requiring immediate surgical intervention.

- Mortality rates are extremely high (**50–80%**, Cleveland Clinic)
- Surgeons must rapidly determine **which bowel is viable vs. ischemic**
  - Costs per hospitalization stay are between $17,500 and $60,000 [2]
- Current assessment methods are **subjective** (visual inspection, palpation, ICG)

Critical Challenge:

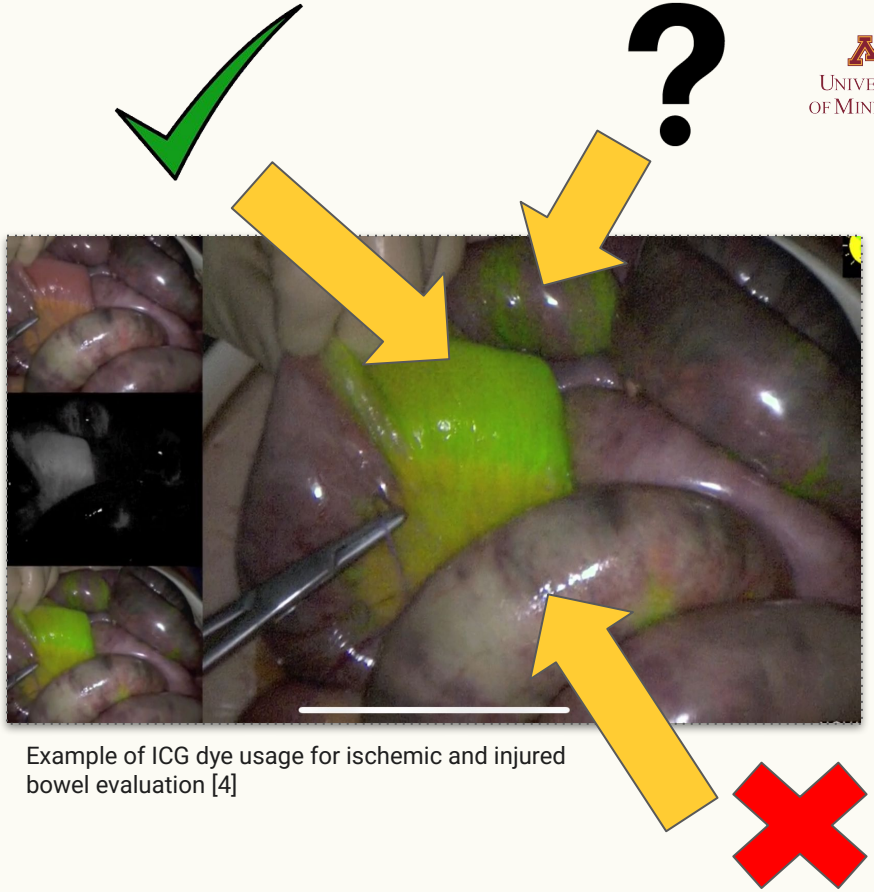
Over-resection → short bowel syndrome, malabsorption, long-term complications

Under-resection → necrotic tissue, sepsis, repeat surgery

Current Limitations

- The current method to identify bowel is to dye the GI system with **Indocyanine green (ICG) dye**.

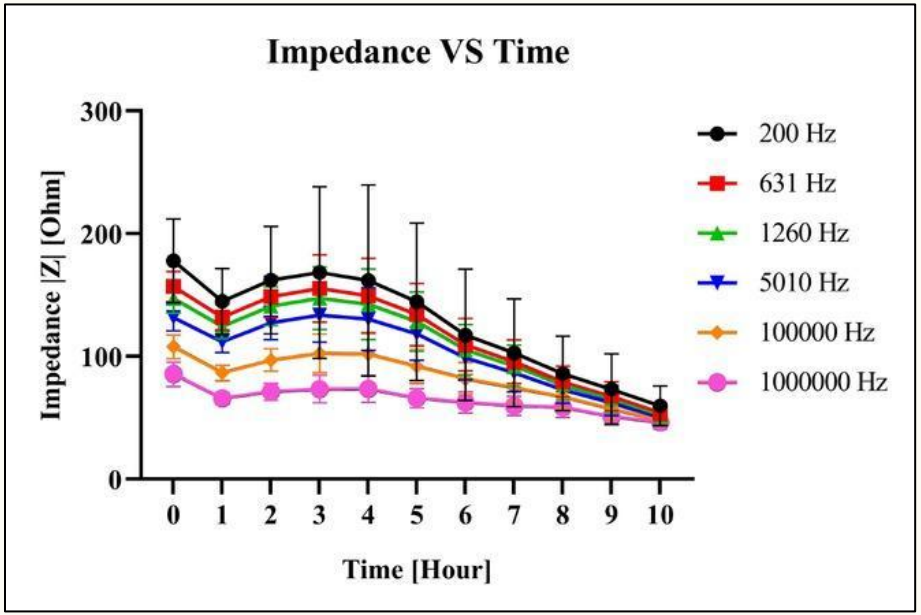
- Studies show that ICG guided surgery in emergency cases only aid **34.6%** of patients [3]



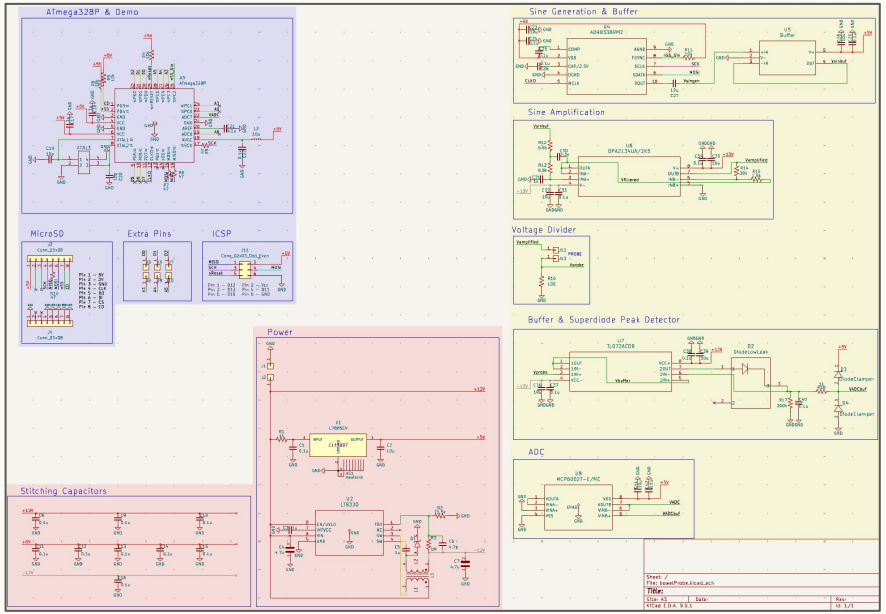
Example of ICG dye usage for ischemic and injured bowel evaluation [4]

There is no reliable, real-time, objective tool to guide intraoperative bowel viability decisions

Working Principles

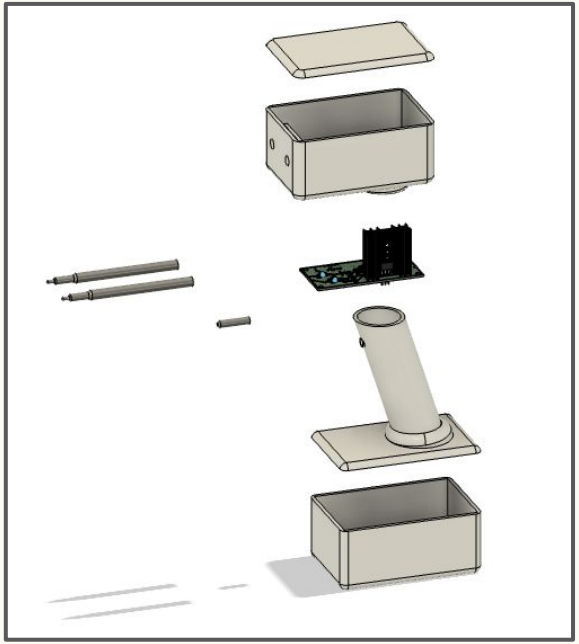


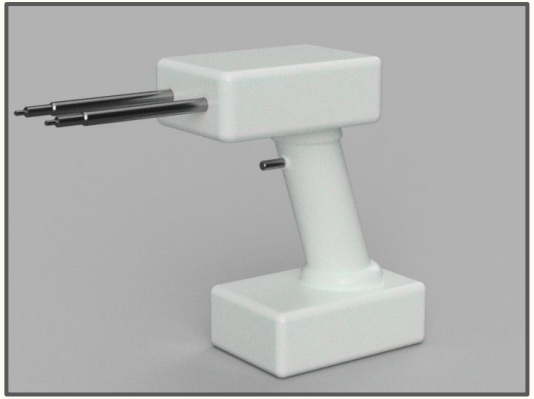
"Plot of mean impedance values with 95 % confidence interval against time (0 to 10 hours) for selected frequencies." [5]

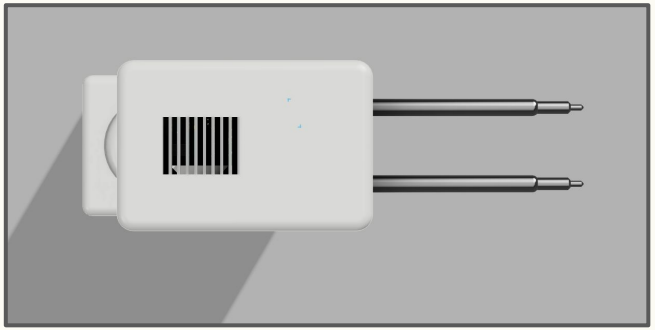


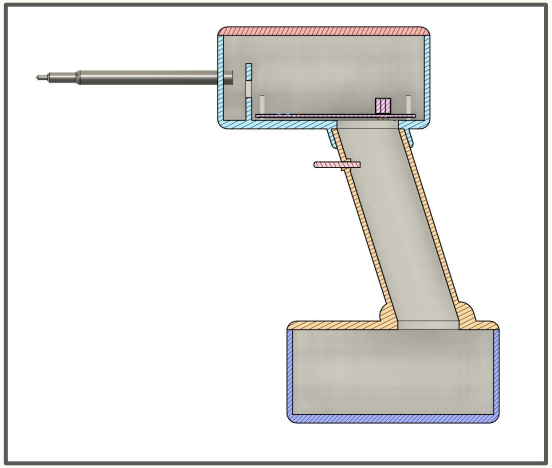
Circuit schematic and architecture for proprietary BowelProbe™

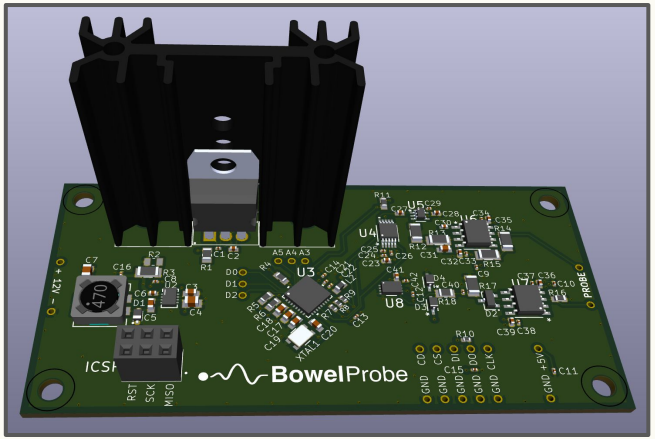
Design









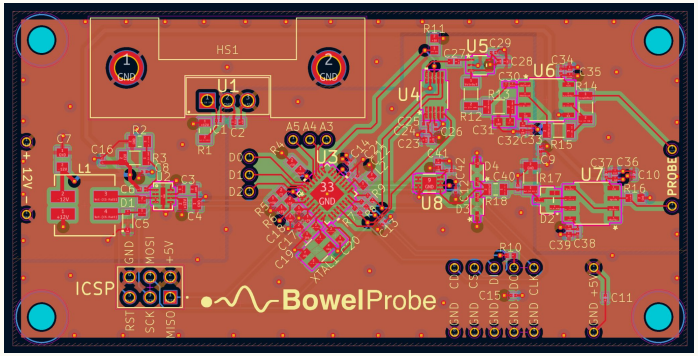


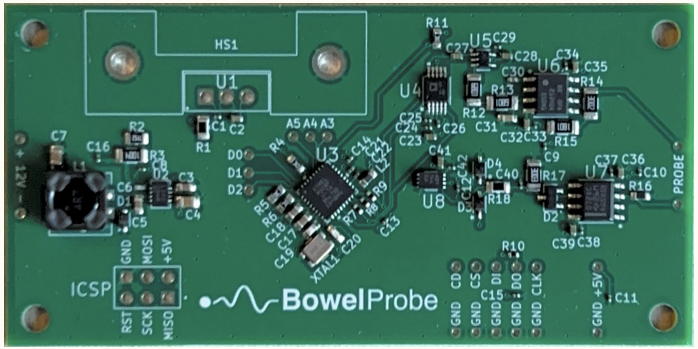
Value Proposition

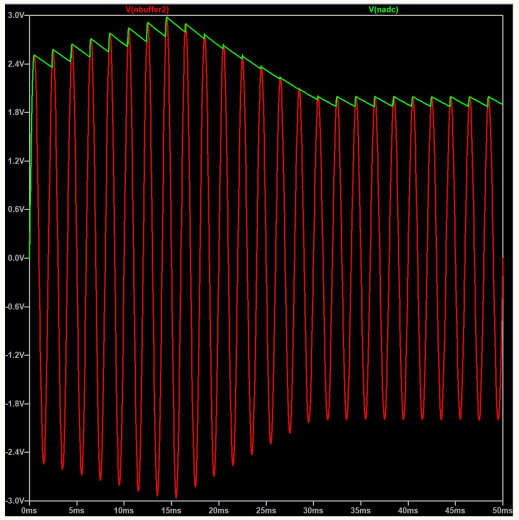
- Portable handheld
- Definitive results
- Instant feedback

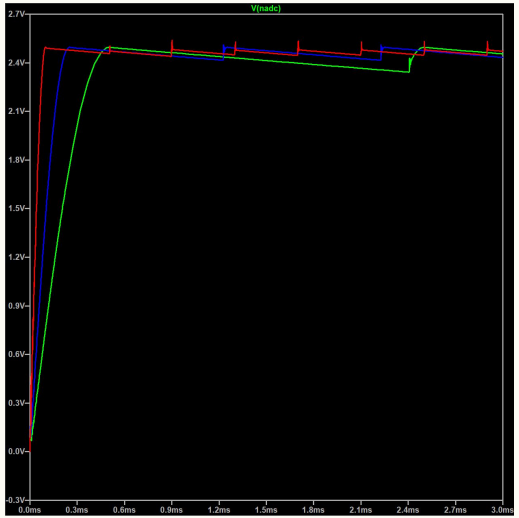
By delivering instant, objective, real time bowel viability assessment to support intraoperative resection decisions, BowelProbe reduces uncertainty, complication risks, and costs.

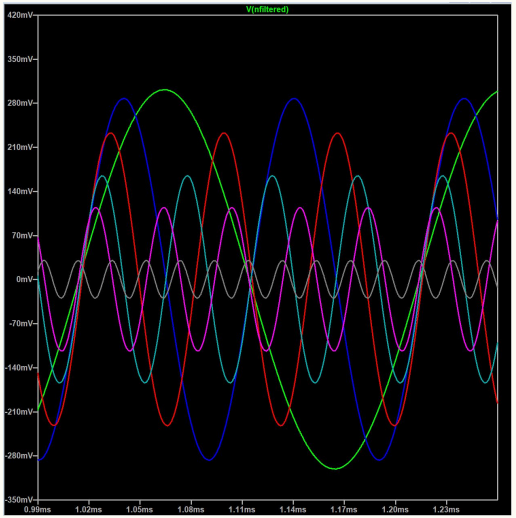
Evidence











5k, 10k, 15k, 20k, 25k, 50k

500, 1k, 2.5k

Intellectual Property

- **System Architecture**
  - Handheld intraoperative device
  - Real-time decision making

- **Measurement Methodology**
  - Multi-frequency electrical impedance spectroscopy

- **Signal Processing**
  - Impedance measurement to viability assessment
  - Instant output

- **Patents**
  - File provisional patent immediate
  - After 12 months file non-provisional patent

- **Trade Secrets**
  - Keep specific algorithms and filtering confidential

Strength of IP Position

- Impedance spectroscopy is not novel but the specific use case and design of BowelProbe is
- Strong IP position due to novel integration of multi-frequency measurement, optimized electrode design, and real-time intraoperative decision support

Product Development Strategy

Multi-phase strategy following ISO 13485

- Gather customer requirements from intended users. Develop design inputs and outputs
- Develop prototypes and initial bench-testing
- Risk Management through ISO 14971
- Formal design verification and validation
  - IEC 60601 Medical Equipment Electrical Safety testing, ISO 10993 Biocompatibility, IEC 62366-1 Usability Testing
  - In-vivo comparative testing using animal models. Clinical Trial not required.

Commercial manufacturing to occur at a US CMO

- Ease of communication and resource management

Development costs prior to commercial launch estimated at $25m.

Months 1–6
Company founded, strategy developed prototypes built
5 Employees | $5M

Months 6–12
Prototypes tested and refined
Additional fundraising
5 Employees | $15M

Months 12–24
Formal V&V
10 Employees | $5M

Months 24–30
FDA submission and approval
15 Employees | $5M

Month 30
Commercialization of device
25 Employees | No further funding

The Market

University of Minnesota

Target Customer
n = 6,000 US Colorectal surgeons
Hospitals and Academic medical centers

Market Size & Growth
Multi-$B surgical guidance market
n = 750k US emergency bowel cases
6–10% annual growth

Key Trends
Real-time intraoperative tools
Adoption of sensing tech
Focus on outcomes & cost reduction

Primary Competitors
ICG fluorescence imaging
Stryker, Medtronic platforms
Emerging sensing tools

Differentiation
Direct electrical activity detection
Real-time audible and visual feedback
Stronger decision accuracy

Barriers to Entry
FDA approval
Surgeon adoption

Pathway to Market

University of Minnesota

Regulatory Pathway

FDA DeNovo Class II submission.

Risks	Mitigations
Product Classification	FDA agreement via Pre-sub with risk-driven data development plan
Inappropriate animal model	Feasibility testing
Implementation of timely Quality Management System	Leveraging established startup systems. Access to experts in industry startup QA and RA systems

Reimbursement

- Fits within existing surgical billing pathways (no new code barrier)
- Preventing one complication offsets device cost → strong hospital incentive
- Established precedent from similar intraoperative technologies → high reimbursement likelihood

Clinical Studies

1. Human clinical trials not needed for regulatory approval.
2. Ex-vivo animal models are sufficient.
3. Comparison to histopathology models

Business Model

1. Product is a reusable capital controller with a single use metal probe.
2. Price will be $30k for the controller, and $500 for each probe. COGs is $2500 for controller and $40 for the probe.
3. Mix of contracted and in-house sales representatives

Management Team

- Board of Directors
  - Three members – One from Investor community, one from scientific community, one medical professional
- Months 1 – 12
  - CEO – 20 years of startup Medical Device experience
  - R&D lead – 15 years of Medical Device experience, successful track record of bringing devices to market
  - RA/QA lead – 15 years of Medical Device experience, successful track record of FDA approval. Knowledge of international regulations
  - R&D Engineer – Strong Electromechanical development background. Software experience.
  - R&D Technician – Strong prototyping experience
- Months 12+
  - Additional R&D, QA, Sales and Marketing personnel hired based on need

BowelProbe is a….

Real-Time, Intra-operative Tool for Quick and Reliable Decision Making

- Intuitive logic and can be adapted for measuring viability of other tissues
- Seamless extension in to emerging robotics assisted surgery (RAS) modes of surgery
- Modular design



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References

[1] N. Kalam-Sakit, I. Obaidat, P. Neupane, M. Khurram EBIR Nazir, and K. Ali Khan, “Endovascular Management of Chronic Mesenteric Ischemia: Technical Success and Long-Term Outcomes Over 10 Years in a UK Tertiary Centre,” *Annals of Vascular Surgery*, vol. 123, Apr. 2026, doi: https://doi.org/10.1016/j.avsg.2025.09.018.

[2] A. NUZZO *et al.*, “320.7: Is there a life after surgery for acute mesenteric ischemia? A retrospective study of 184 survivors,” *Transplantation*, vol. 103, no. 7S2, pp. S18–S19, Jul. 2019, doi: https://doi.org/10.1097/01.tp.0000575512.27316.1c.

[3] B. De Simone *et al.*, “Indocyanine green fluorescence-guided surgery in the emergency setting: the WSES international consensus position paper,” *World Journal of Emergency Surgery*, vol. 20, no. 1, Feb. 2025, doi: https://doi.org/10.1186/s13017-025-00575-w.

[4] Colorectal Disease Journal, “Ischemic and Injured Bowel Evaluation with PINPOINT® Fluorescence Imaging,” *YouTube*, Aug. 13, 2015. https://www.youtube.com/watch?v=XUtpfzV0e1M (accessed Dec. 12, 2024).

[5] J. Hou, R. Strand-Amundsen, S. Hødnebø, T. I. Tønnessen, and J. O. Høgetveit, “Assessing ischemic injury in human intestine ex vivo with electrical impedance spectroscopy,” *Journal of Electrical Bioimpedance*, vol. 12, no. 1, pp. 82–88, Nov. 2021, doi: https://doi.org/10.2478/joeb-2021-0011.

Thank You!

